SAGAR SHRESTHA

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EDUCATION

Ph.D. in Computer Science with AI Minor *Jan 2021 - Expected Feb 2026*
Oregon State University, Corvallis, Oregon, USA GPA : 4.0/4.0

MS in Computer Science *Jan 2021 - March 2023*
Oregon State University, Corvallis, Oregon, USA GPA : 4.0/4.0

Bachelor of Engineering in Electronics and Communications *Oct 2012 - Oct 2016*
Pulchowk Campus, Tribhuvan University, Nepal Distinction

PROFESSIONAL EXPERIENCE

Oregon State University *Jan 2021 – Present*
Graduate Research Assistant (Advisor: Xiao Fu)

- Developed Diversified Flow Matching, for simultaneously transporting diverse set of data distributions; published in ICML [1]
- Pioneered content-style disentangled unsupervised data (e.g., image) translation and generation; applied to natural and OCT medical images, hyperspectral images, single-cell alignment, domain adaptation; published in ICLR and NeurIPS [3, 4, 5].
- Designed performance-guaranteed methods for fundamental problems such as radio map inpainting, resource allocation, and federated multimodal learning by leveraging generative models, GNNs, imitation learning, and nonlinear optimization, CCA; published in TSP and ICASSP [11, 9, 10].
- (Ongoing) Designing novel VLM pretraining objective for noisy large-scale image-text data.

Amazon, WA, US *June 2025 – Oct 2025*
Applied Scientist Intern (Mentors: Penny Guo, Chenchao Shou, MC Nguyen)

- Developed semi-supervised graph transformer model for web-scale heterogeneous graph; improved account takeover detection recall by 7% and precision by 2 % compared to existing production system; Resulted in an inclined/return offer.

Samsung Research America, CA, US *March 2025 – June 2025*
Research Intern, Multimodal Learning (Mentors: Gopal Sharma, Luowei Zhou, Suren Kumar)

- Developed a SOTA fine-tuning-free framework for personalized text-to-image (p-T2I) generation using hypernetworks, and a novel inference method that improves classifier-free guidance in diffusion sampling; submitted [2].

Paaila Technology, Kathmandu, Nepal *Nov 2016 – Sept 2020*
ML Engineer, Team Lead, and Co-founder

- Developed entire navigation stack for waiter robot including sensor fusion, perception, planning, and control; successfully deployed at restaurants and hospitals [Media].
- Other projects: 360 video stitching, Nepali speech recognition and synthesis.

SELECT PUBLICATIONS

Full List: [Google-Scholar]

1. Diversified Flow Matching for Identifiable Translation,
   Sagar Shrestha, Xiao Fu.
   International Conference on Machine Learning (ICML), 2025. [PDF]

2. Fine-tuning Free Personalization of Text-to-Image Generation via Hypernetworks,
   Sagar Shrestha, Gopal Sharma, Luowei Zhou, Suren Kumar.
   (Under Review) 2025.

3. Content-Style Learning from Unaligned Domains: Identifiability under Unknown Latent Dimensions,
   Sagar Shrestha, Xiao Fu.
   International Conference on Learning Representations (ICLR), 2025. [PDF]

4. Identifiable Shared Component Analysis of Unpaired Multimodal Mixtures
   Subash Neupane, **Sagar Shrestha**, and Xiao Fu.
   Neural Information Processing Systems (NeurIPS), 2024. [PDF]

5. Towards Identifiable Unsupervised Domain Translation: A Diversified Distribution Matching Approach,
   Sagar Shrestha, Xiao Fu.
   International Conference on Learning Representations (ICLR), 2024. [PDF]

6. TagSAGE: Account TakeOver Tagging via heterogeneous graph semi-supervised learning,
   Sagar Shrestha, Penny Guo, Chenchao Shou, Manh Cuong Nguyen, Mehran Kafai.
   AMLC Graph Learning Workshop 2025.

7. Unregistered Spectral Image Fusion: Unmixing, Adversarial Learning, and Recoverability,
   Jiahui Song, **Sagar Shrestha**, Xiao Fu.
   (Under Review) *IEEE Transactions on Signal Processing* (TSP), 2025.

8. Translation Identifiability-Guided Unsupervised Cross-Platform Super-Resolution for OCT Images
   Jiahui Song*, **Sagar Shrestha***, Xueshen Li, Yu Gan, and Xiao Fu .
   IEEE SAM, 2024 (*equal contribution) [PDF]

9. Communication-Efficient Federated Linear and Deep Generalized Canonical Correlation Analysis.
   Sagar Shrestha and Xiao Fu.
   IEEE Transactions on Signal Processing (TSP), 2023. [PDF]

10. Deep Generative Model Learning For Blind Spectrum Cartography with NMF-Based Radio Map Disaggregation,
    Sagar Shrestha, Xiao Fu, and Mingyi Hong.
    IEEE ICASSP, 2021. [PDF]

11. Optimal Solutions for Joint Beamforming and Antenna Selection: From Branch and Bound to Graph Neural Imitation Learning,
    Sagar Shrestha, Xiao Fu, and Mingyi Hong.
    IEEE Transactions on Signal Processing (TSP), 2023. [PDF]

12. Quantized Radio Map Estimation Using Tensor and Deep Generative Models,
    Subash Timilsina, **Sagar Shrestha**, and Xiao Fu.
    IEEE Transactions on Signal Processing (TSP), 2023. [PDF]

RELEVANT SKILLS

Machine Learning Areas: Generative models, Graph learning, Multimodal Learning, Self-Supervised & Unsupervised learning, Domain Translation, Causal Representation Learning, Federated Learning, Optimization.

Models & Architectures: Diffusion Models, Flow Matching Models, GANs, VAEs, Transformers, Vision-Language Models (VLMs), Graph Neural Networks (GNNs).

Programming & Tools: Python (PyTorch, JAX, Scikit-learn, Pandas, etc), C/C++, SQL, Git, Bash, Linux, Spark.

Mathematical Foundations: Statistical Learning, Convex & Non-Convex Optimization, Advanced Linear Algebra, Probability & Measure Theory.

TEACHING EXPERIENCE

Substitute Instructor at Oregon State University

- Taught some classes for the grad-level courses: *Matrix Analysis* and *Convex Optimization*

Assistant Lecturer at Thapathali Campus, Nepal *Jan 2019- Jan 2020*

- Taught undergraduate level courses, Instrumentation II and Telecommunications, in ECE department.

REVIEWING

Peer-reviewed more than 30 papers in ICML, NeurIPS, TSP, AAAI, OJSP, ICASSP, Globecom

HONORS

- Ranked 37 out of 15000 applicants in engineering entrance exam. Received merit-based Full scholarship for undergraduate study
- Valedictorian of the class of 2012 at Nobel Academy (high school level study in Nepal).
- First Position in annual National Technological Festival LOCUS for three years in a row during my undergraduate study for software and hardware projects - 2014, 2015, 2016
- Four awards in international robot competition ABU robocon 2015 and 2016 as a team member of team Nepal [Awards]
- Awards for my startup: Most Creative Business Nepal (CBC Cup 2018), Best Startup Award (ICT Award 2017), Most Innovative Product Award (ICT Award 2017), Best Industry Technology Award (FNCCI 2017) [Website]